

Effect of ZnO Nanoparticles on the Surface Potential Decay of Epoxy Nanocomposites

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Abstract – Epoxy resin as a typical electrical insulation material has been widely applied in the power equipment. However, the electrical properties of epoxy resin are strongly affected by the surface charge accumulation. One way to solve the surface charge accumulation may be to use the nonlinear resistive field grading nanocomposites. In this paper, we aim to study the effect of ZnO nanoparticles on the surface potential decay (SPD) and surface charge accumulation of epoxy nanocomposites under negative corona charging. Samples were prepared by dispersing the ZnO nanoparticles into epoxy matrix with weight percentages of 0, 1, 3, 7 and 10, respectively. The SPD measurements were performed on these samples charged by negative corona effect. The results show that both the surface potential decay and the decay rate depend on the filler loading. Such dependences are closely related to the interaction zone of nanoparticles through analyzing the trap distribution.

Keywords: epoxy nanocomposite, ZnO, surface potential decay, trap distribution, interaction zone

I. INTRODUCTION

Epoxy resin and its composites have been widely used in a variety of electrical applications, such as bushings and gas-insulated switchgear spacers, due to their excellent electrical and mechanical properties [1]. Comparing with epoxy microcomposites, the electrical properties, e.g. conductivity, permittivity, dielectric loss and breakdown strength, of epoxy nanocomposites have been improved to some degree [2]. The surface charges accumulated on the epoxy resin and its composites usually have a great impact on the electrical properties. The accumulated surface charges may lead to the electric field distortion near the epoxy insulators and even cause an unexpected flashover [3]. Therefore, it is necessary to study the surface charge behavior of pure epoxy resin as well as its composites.

The SPD measurement is very convenient to characterize the surface charge behavior of insulating materials, and can be used to estimate the trap distribution without considering any previous physical interpretation of the SPD curves. In addition, the SPD measurement can also be used to evaluate the carrier mobility and the bulk conductivity under proper conditions [4]. With the help of the SPD measurement, many studies have been performed on the influence of nanoparticles, such as titanium dioxide (TiO₂) and silica (SiO₂), on the surface charge accumulation and the decay behaviors of epoxy nanocomposites [5, 6]. The SPD results show that the charge accumulation and the decay rate depends on the filler

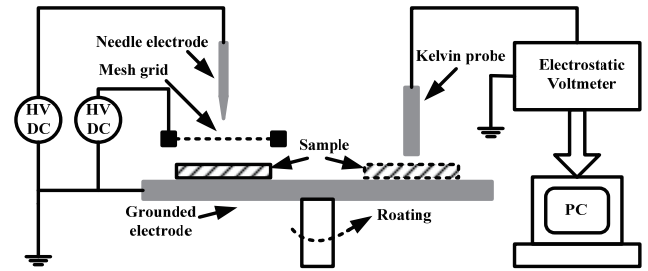


Figure 1. Schematic arrangement of the SPD measurement setup.

loading, and the decay rate nearly decrease at first but then increase when the filler content is increased from 0 to 7 wt%. Moreover, the decay rate of pure epoxy is always larger than that of epoxy nanocomposites, thus these nanocomposites may inhibit the dissipation of the accumulated charges.

To relieve the charge accumulation, the nonlinear resistive field grading composites can be used and can be prepared by filling appropriate filler content of semiconducting fillers, such as silicon carbide (SiC) and zinc oxide (ZnO), into the polymer matrix [7, 8]. These grading composites can increase a little conductivity to dissipate the accumulated charges and then recover to the insulating state. However, very few works have been focused on the surface charge accumulation and the decay behaviors of epoxy based nonlinear resistive field grading composites, such as epoxy/ZnO nanocomposite.

In this paper, the SPD and surface charge accumulation of epoxy/ZnO nanocomposites are studied by SPD measurement. The results show that the SPD behavior and the decay rate are greatly dependent on the filler loading. The trap distributions are also obtained by the SPD measured data.

II. EXPERIMENTAL DETAILS

The experimental samples were prepared by dispersing the ZnO nanoparticles with an average particle size of 40 nm into epoxy matrix. The filler loading was 1, 3, 7 and 10 wt%, respectively. For comparison, the pure epoxy resin was also prepared with bisphenol-A epoxy resin (WSR618 E-51) as a matrix and methyltetrahydrophthalic anhydride (GH-9303) as a hardener. The thickness of samples is about 0.23 mm.

The SPD experimental setup as shown in Fig. 1 consists of a corona charge system and a measurement system. The corona charge system including a high-voltage needle electrode, a mesh grid and a grounded electrode is used to charge the sample's surface with ions of either sign. Whereas, the measurement system involving a Kelvin type probe, a noncontact electrostatic voltmeter and a computer is applied to measure and record the SPD values of the charged sample.

This work was supported by the National Science Foundation of China (NSFC) under Projects with No. 51337008, National Basic Research Program of China with No. 2015CB251003, and NSFC with Nos. 11275146 51221005, 51323012, and 51507124.

The sample was charged for 3 min by the corona charge system of the SPD experimental setup. The needle electrode and the mesh grid were applied by dc high voltage of -17 kV and -5 kV, respectively. The surface potential values of the charged sample was measured and recorded for 3.5 h through the measurement system of the ISPD experimental setup. By repeating the same procedure, the surface potential values of all epoxy nanocomposites were obtained, respectively. Such experiments were performed at room temperature with relative humidity of about 45%.

III. ANALYSIS OF SURFACE POTENTIAL DECAY

The SPD of the charged polymeric surface may be related to the surface conduction, gas neutralization and bulk transport. According to the electrode arrangement and sample dimension, the electric field between the charged surface and the grounded electrode is higher than that along the surface, so the effect of the surface conduction can be neglected. In addition, as the Kelvin probe has the same potential with the measured polymeric surface, the gas neutralization can hardly affects the SPD. Therefore, the bulk transport of the charge carriers is thought to be the main decay mechanism in this work.

A decay rate of SPD is introduced [5], defined as follows,

$$D = (U(t_0) - U(t_i)) / U(t_0) \times 100\%. \quad (1)$$

where D is the decay rate, $U(t_0)$ and $U(t_i)$ are the surface potentials at the initial decay time t_0 and at the last decay time t_i , respectively.

The trapping and de-trapping of charge carriers in the bulk are associated with the trap distribution which can be characterized by the energy level and the energy density. The energy level of traps is related to the decay time, and the relationship between the energy level of traps and the decay time can be defined as follows [9],

$$E_T(t) = kT \ln(vt). \quad (2)$$

where E_T is the energy level of traps, k is the Boltzmann's constant, T is the Kelvin temperature, v is the attempt to escape frequency, and t is the decay time.

The energy density of traps can be denoted as [9],

$$N_T(t) = t \frac{\epsilon_0 \epsilon_r}{qd} \frac{dU(t)}{dt} \quad (3)$$

where ϵ_0 and ϵ_r are the vacuum dielectric constant and the relative dielectric constant, respectively. q is the elementary charge. d is the sample's thickness. $U(t)$ is the measured surface potential.

IV. RESULTS AND DISCUSSIONS

A. Surface Potential Decay

After charging the sample, the charges accumulated on the sample's surface experience the bulk transport and transfer from the surface to the grounded electrode, so the surface potential decays with time. Fig. 2 illustrates the measured surface potential varying with the decay time for all samples.

For each sample, the surface potential decays quickly during the initial time but decays slowly with the elapse of the

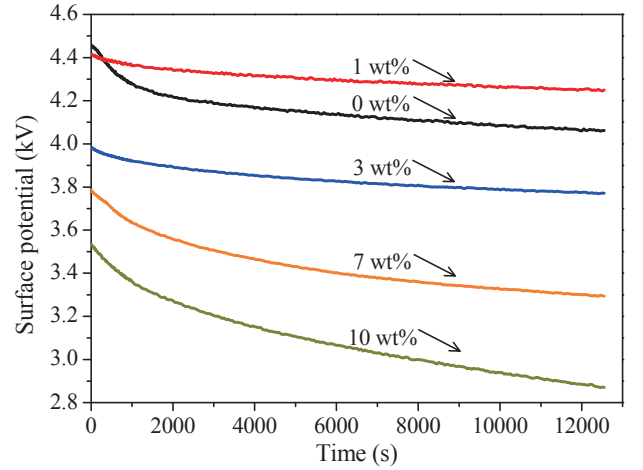


Figure 2. Surface potential varying with the decay time.

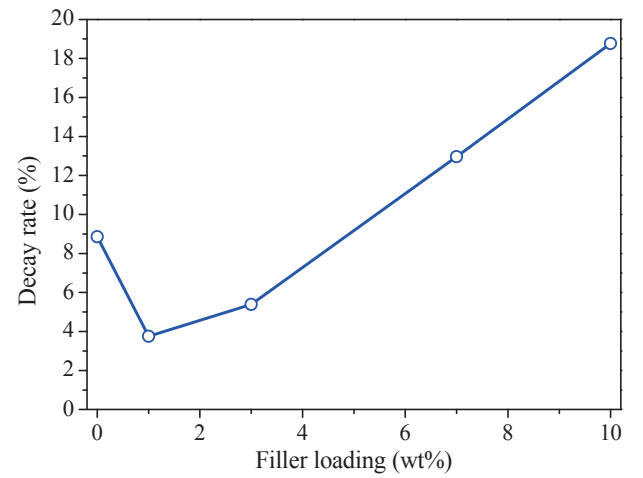


Figure 3. Relationship between the decay rate and the filler loading.

decay time. Although the surface potential characteristics of all samples have similar tendency, the initial surface potential values of all samples are different from each other. It seems that the initial surface potential value decreases with the increasing filler loading. Moreover, the surface potential characteristics of all samples are distinct from each other. Consequently, the SPD of epoxy/ZnO nanocomposites under negative corona charging depends on the filler loading.

B. Decay rate

To better understand the SPD characteristic, the decay rate of each sample is obtained according to (1). Fig. 3 shows the relationship between the decay rate and the filler loading. The results shows that, under the same charging level, the decay rate is decreased at first but is then increased when the filler loading is increased from 0 wt% to 10 wt%. Compared with the results reported in [5, 6], the decay rate of pure epoxy resin, is not always larger than that of epoxy nanocomposites in this study.

The decay rate of 0 wt% sample is larger than that of the slight filler loading samples, but is smaller than that of the high filler loading samples. In the whole variation of the filler loading as shown in Fig. 3, the decay rate of 1 wt% sample is the smallest whereas that of 10 wt% sample is the largest. Hence, it is suggested that the decay rate of the surface potential characteristic of epoxy/ZnO nanocomposites can be

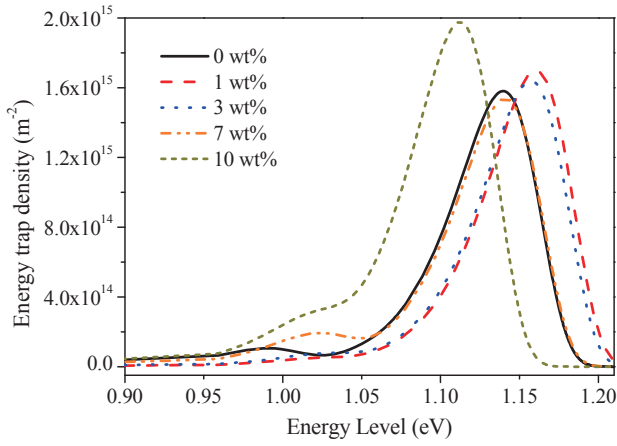


Figure 4. Trap distribution of epoxy/ZnO nanocomposites.

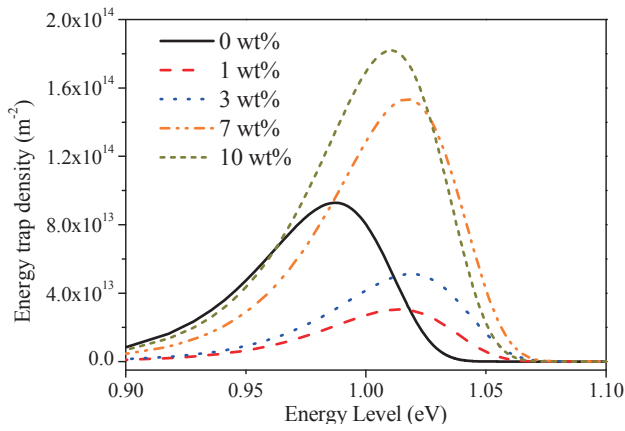


Figure 5. Shallow trap distributions of epoxy/ZnO nanocomposites.

regulated and controlled by the filler loading.

C. Trap Distribution

The trap distributions of pure epoxy and epoxy/ZnO nanocomposites are obtained based on (2) and (3) as well as the surface potential characteristics as shown in Fig.2. The relative dielectric constant used in (3) for each sample is 4.24, 4.44, 4.82, 4.96, 5.09 and 6.96, respectively, which are measured by a Novocontrol broadband dielectric spectrometer at room temperature.

Fig. 4 shows the trap distributions of all samples used in this work, where the energy level of traps mainly ranges from 0.90 to 1.20 eV. The profiles of the trap distributions shown in Fig. 4 are similar, and the trap distribution of each sample seems have two peaks, one peak locating at 0.90~1.05 eV whereas the other peak locating at 1.05~1.20 eV.

The peak separation is performed to further analyze the trap distribution [9]. The shallow and deep distributions are shown in Fig. 5 and Fig. 6, respectively.

For the shallow traps, it can be found in Fig. 5 that the energy level is mainly located at 0.90~1.05 eV. In addition, the energy level corresponding to the peak energy density of shallow traps ranges from 0.99 to 1.02 eV when the filler loading is increased from 0 to 10 wt%, so the filler loading has little effect on the energy level of shallow trap distribution. It can also be seen from Fig. 5 that the energy density of shallow traps, specifically the peak energy density, is decreased from 9.28×10^{13} to $3.04 \times 10^{13} \text{ m}^{-2}$ when the filler

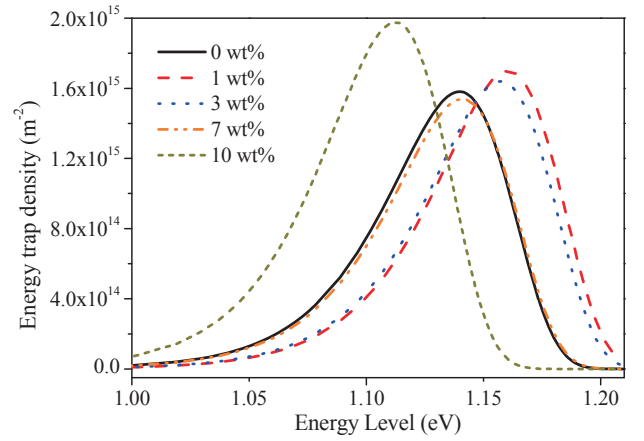


Figure 6. Deep trap distributions of epoxy/ZnO nanocomposites.

loading is increased up to 1 wt%, but it is then increased to $1.82 \times 10^{14} \text{ m}^{-2}$ when the filler loading is increased from 1 to 10 wt%. The energy density of shallow traps seems have a similar tendency with the decay rate, namely decreasing at first and then increasing with the increase in the filler loading.

For the deep traps, as shown in Fig. 6, the energy level corresponding to the peak energy density is increased from 1.14 to 1.16 eV when the filler loading is increased from 0 to 1 wt%, but it is then decreased from 1.16 to 1.11 eV when the filler loading is increased from 1 to 10 wt%, thus the energy level of deep traps is nearly increased at first but is then decreased with the increasing filler loading, which has an opposite tendency with the decay rate. In comparison, the peak energy density ranges from $1.54 \times 10^{15} \text{ m}^{-2}$ to $1.98 \times 10^{15} \text{ m}^{-2}$ and changes a little when the filler loading is increased up to 10 wt%, so the filler loading can barely affect the energy density of deep trap distribution.

As a result, the energy density of shallow trap distribution and the energy level of deep trap distribution contribute to the decay rate of the surface potential.

Based on the multi-region structure [10], the interaction zone of nanoparticles has a significant impact on the electrical properties, such as the decay rate of the surface potential, of epoxy nanocomposites. For slight filler loading (eg. 1 wt%), the nanoparticle can be regarded as an isolated particle and there exist plenty of deep traps in the interaction zone of the isolated nanoparticles, so the trapped charges are seriously restrained by these traps and the decay rate is decreased. For low filler loading (eg. 3 wt%), the interaction zone of the nanoparticles may partly overlap and the thickness of the interaction zone (especially the transition region) is increased, so the trapped charges can get enough energy from electric field to detrapp and the decay rate is increased. For high filler loading (eg. 10 wt%), the percolation phenomenon may occur and many conductive paths are formed through the overlap of the transition region in the bulk of nanocomposite, so the trapped charges are easier to detrapp and the decay rate is increased further. In addition, the semi-conduction of ZnO nanoparticles may also contribute to the increasing decay rate.

V. CONCLUSIONS

Effect of ZnO nanoparticles on the SPD behavior and surface charge accumulation of epoxy nanocomposites is studied by SPD measurements. The important conclusions are summarized as follows.

1. The SPD characteristics of epoxy/ZnO nanocomposites have a similar time variation, but are also distinct from each other due to the difference in surface potential values.

2. The SPD decay rate is decreased at first but is then increased with the increase in filler loading.

3. According to the trap distribution, the decay rate of the surface potential of epoxy/ZnO nanocomposite is closely related to the energy density of shallow trap distribution and the energy level of deep trap distribution.

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